

Here are concise meanings (column-level unless noted), with units and sign conventions where applicable:

- `ch4_varsgrnd_ch4_cond_col`: Ground-to-atmosphere methane conductance over lakes [m s^{-1}]. 0 when surface ice is present ($> \sim 0.1$); otherwise $1/(\text{lakeresist} + \text{lake_raw})$.
- `col_efeflx_snomelt`: Snow-melt heat flux [W m^{-2}] (positive = energy used to melt snow).
- `col_eferrsoi`: Soil/lake energy conservation residual after step [W m^{-2}] (small errors are folded into sensible heat).
- `col_efimelt`: Snow layer phase-change flag (0=none, 1=melting, 2=freezing) per layer.
- `col_eshc_soi`: Soil heat content only [MJ m^{-2}] (excludes snow; accumulated from $\text{cv} \cdot T$ over soil layers).
- `col_eshc_soisno`: Combined soil+snow+lake heat content [MJ m^{-2}].
- `col_est_lake`: Lake-layer temperatures [K] (1.. nlevlak).
- `col_est_soisno`: Soil/snow layer temperatures [K] ($-\text{nlevsno}+1..\text{nlevgrnd}$).
- `col_wfqflx_snofrz_lyr`: Snow freezing rate per snow layer [$\text{kg m}^{-2} \text{s}^{-1}$] (positive definite).
- `col_wfqflx_snomelt`: Snow melt water flux [$\text{mm H}_2\text{O s}^{-1}$].
- `col_wfqflx_snow_melt`: Net snow-melt rate aggregated over column [$\text{mm H}_2\text{O s}^{-1}$].
- `col_wsh2osno`: Snow water equivalent (SWE) on column [$\text{mm H}_2\text{O}$].
- `col_wssnow_depth`: Snow depth [m].
- `lakestate_varsbetaprime_col`: Effective shortwave “beta” for surface absorption (NIR fraction; with snow it becomes `sabg_lyr` at top snow layer divided by `sabg`) [dimensionless].
- `lakestate_varslake_icefrac_col`: Lake-layer ice mass fraction [0–1].
- `lakestate_varslake_icethick_col`: Column-integrated physical lake ice thickness [m].
- `lakestate_varslakeresist_col`: Lake water transport resistance used for CH_4 pathway [s m^{-1}].
- `lakestate_varssavedtke1_col`: Eddy “conductivity” at top lake level saved for next step [$\text{W m}^{-1} \text{K}^{-1}$] (= $\text{kme}(1) \cdot \text{cwat}$ when unfrozen).
- `veg_efeflx_gnet`: Net heat flux into ground at surface interface [W m^{-2}] (positive into ground).
- `veg_efeflx_sh_grnd`: Sensible heat flux from ground to atmosphere [W m^{-2}] (positive to atmosphere).
- `veg_efeflx_sh_tot`: Total sensible heat flux (all surface components) [W m^{-2}] (positive to atmosphere).
- `veg_efeflx_soil_grnd`: Heat flux into snow/lake (ground heat flux) [W m^{-2}] (positive into column).

Here’s a concise validation guide for each output: shape, units, expected range, and notes/sign.

- `ch4_varsgrnd_ch4_cond_col`
 - shape: (ncol)

- units: m s^{-1}
- range: 0 to $\sim 1\text{e-}2$ (typical $0\text{--}1\text{e-}4$)
- notes: 0 if $\text{lake_icefrac}(\text{top}) > 0.1$
- *col_efeflx_snomelt*
 - shape: (ncol)
 - units: W m^{-2}
 - range: ≥ 0 (typically $0\text{--}500$)
 - notes: energy used to melt snow (positive-definite)
- *col_eferrsoi*
 - shape: (ncol)
 - units: W m^{-2}
 - range: near 0 after correction ($|\text{err}| < 1\text{e-}6$ to $1\text{e-}3$)
 - notes: small residuals are folded into sensible heat
- *col_efimelt*
 - shape: (ncol, $-\text{nlevsno}+1:0$) [snow layers only]
 - units: flag
 - range: $\{0,1,2\}$
 - notes: 0=none, 1=melting, 2=freezing
- *col_eshc_soi*
 - shape: (ncol)
 - units: MJ m^{-2}
 - range: ≥ 0 (typical tens to thousands)
 - notes: soil-only heat content
- *col_eshc_soisno*
 - shape: (ncol)
 - units: MJ m^{-2}
 - range: ≥ 0 (typical hundreds to several thousand)
 - notes: soil+snow+lake combined
- *col_est_lake*
 - shape: (ncol, $1:\text{nlevlak}$)
 - units: K
 - range: $\sim 240\text{--}320$ (typical $260\text{--}305$)
 - notes: bounded around freezing/overtake physics
- *col_est_soisno*
 - shape: (ncol, $-\text{nlevsno}+1:\text{nlevgrnd}$)
 - units: K
 - range: $\sim 230\text{--}320$ (typical $255\text{--}305$)
 - notes: spans snow (≤ 0) and soil (≥ 1) indices
- *col_wfqlx_snofrz_lyr*
 - shape: (ncol, $-\text{nlevsno}+1:0$)
 - units: $\text{kg m}^{-2} \text{s}^{-1}$
 - range: ≥ 0

- notes: positive-definite freezing rate per snow layer
- *col_wfqflx_snomelt*
 - shape: (ncol)
 - units: mm H₂O s⁻¹
 - range: ≥ 0 (typically 0–1e-4)
 - notes: snow melt water flux
- *col_wfqflx_snow_melt*
 - shape: (ncol)
 - units: mm H₂O s⁻¹
 - range: ≥ 0 (typically 0–1e-4)
 - notes: column net snow melt
- *col_wsh2osno*
 - shape: (ncol)
 - units: mm H₂O
 - range: ≥ 0 (site: 0–1000+)
 - notes: snow water equivalent
- *col_wssnow_depth*
 - shape: (ncol)
 - units: m
 - range: ≥ 0 (site: 0–5+)
 - notes: physical snow depth
- *lakestate_varsbetaprime_col*
 - shape: (ncol)
 - units: —
 - range: 0–1
 - notes: effective NIR fraction; with snow, sabg_lyr(top)/sabg
- *lakestate_varslake_icefrac_col*
 - shape: (ncol, 1:nlevlak)
 - units: —
 - range: 0–1
 - notes: layer ice mass fraction
- *lakestate_varslake_icethick_col*
 - shape: (ncol)
 - units: m
 - range: ≥ 0 (typical 0–3+)
 - notes: integrated physical ice thickness
- *lakestate_varslakeresist_col*
 - shape: (ncol)
 - units: s m⁻¹
 - range: ≥ 0 (typical 1e4–1e8)
 - notes: sum dz/kme over non-convective lake layers
- *lakestate_varssavedtke1_col*

- shape: (ncol)
- units: $\text{W m}^{-1} \text{K}^{-1}$
- range: ≥ 0 (typical $\sim 0\text{--}10$)
- notes: $\text{kme}(1) \cdot \text{cwat}$ when unfrozen
- *veg_efeflx_gnet*
 - shape: (npft) or (npatch) depending on layout; here accessed by p
 - units: W m^{-2}
 - range: roughly -500 to $+500$ (can exceed in extremes)
 - notes: positive into ground
- *veg_efeflx_sh_grnd*
 - shape: (npft)/(npatch)
 - units: W m^{-2}
 - range: roughly -500 to $+500$
 - notes: positive to atmosphere
- *veg_efeflx_sh_tot*
 - shape: (npft)/(npatch)
 - units: W m^{-2}
 - range: roughly -500 to $+500$
 - notes: total sensible heat (all components)
- *veg_efeflx_soil_grnd*
 - shape: (npft)/(npatch)
 - units: W m^{-2}
 - range: roughly -500 to $+500$
 - notes: positive into column (snow/lake)

If you want, I can draft assert-style checks (bounds, monotonicity for thickness sums, finite values) you can plug into a unit test harness.

Here's a tight checklist of high-value unit-test assertions for LakeTemperature outputs. Use small tolerances (e.g., $1\text{e-}9$ to $1\text{e-}6$) as noted.

- Dimensions/indices
 - *col_est_lake*: shape (ncol, 1:nlevlak)
 - *col_est_soisno*: shape (ncol, -nlevsno+1:nlevgrnd)
 - *lakestate_varslake_icefrac_col*: shape (ncol, 1:nlevlak)
 - Snow-layer fields (*col_efimelt*, *col_wf_qflx_snofrz_lyr*): defined only on $-\text{nlevsno}+1..0$
- Finite values
 - All temperatures, fluxes, and resistances are finite (no NaN/Inf): *t_lake*, *t_soisno*, *eflx_** (*gnet*, *sh_grnd*, *sh_tot*, *soil_grnd*), *eflx_snomelt*, *qflx_** (*snofrz_lyr*, *snomelt*, *snow_melt*), *h2osno*, *snow_depth*, *lakeresist*, *savedtke1*, *grnd_ch4_cond* (if use_lch4)
- Physical bounds
 - *lakestate_varslake_icefrac_col* $\in [0, 1]$ (all layers)
 - *lakestate_varsbetaprime_col* $\in [0, 1]$

- $col_wsh2osno \geq 0$; $col_ws_snow_depth \geq 0$
- $col_wfqflx_snofrz_lyr \geq 0$ (each snow layer)
- $col_wfqflx_snomelt \geq 0$; $col_wf_qflx_snow_melt \geq 0$
- $col_efeflx_snomelt \geq 0$
- $lakestate_varslakeresist_col \geq 0$; $lakestate_vars_savedtke1_col \geq 0$
- $col_eshc_soi \geq 0$; $col_es_hc_soisno \geq 0$ and $hc_soisno \geq hc_soi$
- Aggregation/consistency
 - $sum_j\ qflx_snofrz_lyr(:,j) == col_wfqflx_snofrz_col$ (abs diff $\leq 1e-10$)
 - $lake_icethick_col == \sum_j\ lake_icefrac_col(:,j) \cdot dz_lake(:,j) \cdot denh2o/denice$ (abs diff $\leq 1e-9$)
 - $col_efimelt \in \{0,1,2\}$ per snow layer (0=none, 1=melting, 2=freezing)
- Energy residual handling
 - $col_eferrsoi$ is driven to 0 after incorporation, or $|errsoi| \leq 1e-6$ (post-correction)
- CH4 conductance behavior (when $use_lch4 = .true.$)
 - If $lake_icefrac_col(:,1) > 0.1$ then $ch4_varsgrnd_ch4_cond_col == 0$
 - Else $ch4_varsgrnd_ch4_cond_col \geq 0$ and finite
- Flux sign conventions (spot checks)
 - veg_efeflx_gnet : positive = into ground (value finite; allow either sign depending on case)
 - $veg_efeflx_soil_grnd$: positive = into column (finite)
 - $veg_efeflx_sh_grnd$ and $veg_ef_eflx_sh_tot$: positive = to atmosphere (finite)
- Optional physics sanity (tighter checks if you want)
 - Where $lake_icefrac_col(:,j) \approx 1$, enforce $|t_lake(:,j) - tfrz| \leq 1e-3$
 - $savetke1_col > 0$ only when surface is unfrozen ($t_lake(:,1) > tfrz$ and $snl == 0$)

These assertions cover shapes, finiteness, physical bounds, mass/energy-consistent sums, CH4 gating under ice, and sign conventions.